

20240610 EB_MN (TDP43, qPCR)

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Status: In progress
Entry Created On: 25 Jun 2024 23:18:55 UTC
Entry Last Modified: 18 Jul 2024 20:23:38 UTC
Export Generated On: 01 Sep 2026 18:23:56 UTC

- 2024年6月10日 星期一
- split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs
- 2024年6月12日 星期三
- T2 -> T3
- 2024年6月14日 星期五
- T3 -> T4 (~14 mL for over-weekend)
- 2024年6月17日 星期一
- T4 -> T5
- 2024年6月19日 星期三
- T5 -> T6
- 2024年6月21日 星期五
- T6 -> T6 (~14mL for over-weekend)
- 2024年6月24日 星期一
- T6 -> T7
- Dissociation
1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs

2. Remove the PLO and wash the plate with PBS once

3. Coat the plate with laminine for 1hr.

4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).

5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.

6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)

7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)

8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.

9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

Table1								
	A	B	C	D	E	F	G	H
1	cell	passage	trypsin	HIFBS/CTWM	cell count (M/mL)	volume	total	per-well (K)
2	WT11 (Anouk)	P32	5.5	11	1	6	6	333.333
3	WT11 (YST)	P34						
4	iPSC5319	PX+12	4+0.5(1st round)	9	3.35	2	6.7	
5	iPSC4517	P12+11	4+0.5(1st round)	9	0.408	2	0.816	272

WT-11 For Anouk				
	1	2	3	4
A	+vLH92 (4uL/well)			
B				
C				

PLATE A WT-11 (for TDP-43 localization)				
	1	2	3	4
A				
B				
C				

PLATE B RNA extraction				
	1	2	3	4
A	610K Ctrl	305K Ctrl	305K Ctrl	272K Mut
B				
C				

2024年6月26日 星期三

T7-> BP T8

2024年6月28日 星期五

400uL BP T8 -> 500 uL BP T8

2024年7月1日 星期一

400uL BP T8 -> 500 uL BP T8

2024年7月2日 星期二

PLATE B

treat row A with 700uL hypo media at 5:35 pm

2024年7月3日 星期三

RNA

PLATE B

- 1. treat row B with 600uL hypo media at 4:30 pm
- 2. wash and freeze the cells at 5:30 pm
- 3. store the cells in -80C

Imaging

PLATE A

Treat SirDNA: (1:6000): remove 400uL and add 200uL BPT8 with SirDNA (600uL in total per well)

Drug prep: 100uL BP T8 with drug

Table2										
	A	B	C	D	E	F	G	H	I	J
1		stock	unit	working	unit	dilution	total media	drug added	total drug added	added media volume
2	Importazole	100.0	mM	40.0	uM	2500.0	1600.0	0.6	#REF!	225.0
3	Verdinexor	50.0	uM	50.0	nM	1000.0	1600.0	1.6	6.4	225.0
4				150.0	nM	333.3	1600.0	4.8		take 20 uL from above and add 185 uL T8

Run every thing all together with ND sequencing acquisition

Setting: New spinning disk confocal, 600x900, 16bit, HDR, 15 min in total

O.C. ^

	A	B	C	D
1	channel	bit	power (%)	exposure time
2	640	16	10	100 ms
3	561	16	20	200 ms
4	488	16	50	1 s

PLATE A WT-11 (for TDP-43 localization)1 ^

	1	2	3	4
A	1	2		
B	6	3		
C	5	4		

2024年7月18日 星期四

RNA extraction with [PureLink™ RNA Mini Kit \(quick protocol\)](#)

Lysis and Homogenization

- 1. Prepare lysis buffer: 3000uL lysis buffer + 30 uL 2-mercaptoethanol (100x)
- 2. Take the plate out and plate it on ice
- 3. Add 300uL lysis buffer for each well and plate the whole ice bucket on the shaker for 15min
- 4. Collect the lysis buffer into 1.5mL tubes
- 5. Combine column 1 - 3

PLATE B RNA extraction1 ^

	1	2	3	4
A				
B	610K	305K	305K	272K
C	Ctrl	Ctrl	Ctrl	Mut

- 6. Homogenization: Pass the lysate 5–10 times through an 18- to 21-gauge syringe needle.

RNA Purification

- 1. Add one volume 70% ethanol to each volume of cell homogenate.
 - 900 uL for 5319

- 300 uL for 4517
- 2. Vortex to mix thoroughly and to disperse any visible precipitate that may form after adding ethanol.
- 3. Transfer up to 700 µL of the sample (including any remaining precipitate) to the spin cartridge (with the collection tube).
- 4. Centrifuge at 12,000 × g for 30 seconds at room temperature. Discard the flow-through, and reinsert the spin cartridge into the same collection tube.
- 5. Repeat Steps 3–4 until the entire sample has been processed.
- 6. Add 350 µL Wash Buffer I to the spin cartridge.
- 7. Centrifuge at 12,000 × g for 30 seconds at room temperature. Discard the flow-through.
- 8. Prepare DNaseI (10uL DNase I + 70 uL water, 80uL in total per rxn). Add 80 uL DNase I into the column and incubate for 15 min at RT.
- 9. Add 350 Wash Buffer I to the spin cartridge and centrifuge at 12,000 × g for 30 seconds at room temperature. Discard the flow-through and the collection tube. Place the spin cartridge into a new collection tube.
- 10. Add 500 µL Wash Buffer II with ethanol to the spin cartridge.
- 11. Centrifuge at 12,000 × g for 30 seconds at room temperature. Discard the flow-through.
- 12. Repeat Steps 8–9 once.
- 13. Centrifuge the spin cartridge at 12,000 × g for 1–2 minutes to dry the membrane with bound RNA. Discard the collection tube and insert the spin cartridge into a recovery tube.
- 14. Add 30–100 µL (50uL) RNase–free water to the center of the spin cartridge.
- 15. Incubate at room temperature for 1 minute.
- 16. Centrifuge the spin cartridge for 2 minutes at ≥**12,000** × g at room temperature to elute the RNA from the membrane into the recovery tube. Note: If the expected RNA yield is >100 µg, perform 3 sequential elutions of 100 µL each. Collect the eluates in a single tube.
- 17. Store your purified RNA in -80 freezer.

RT with Thermo Scientific Maxima™ H Minus cDNA Synthesis Master Mix with dsDNase

Performed together with 0603MN

1. Add the following reagents into a sterile, RNase free tube on ice in the following order:
(make total volumn into 12 uL)

 image.png

10X dsDNase Buffer	1 μ L
dsDNase	1 μ L
Template RNA • Total RNA or • poly (A) mRNA or • Specific RNA	1 pg to 5 μ g 0.1 pg to 500 ng 0.01 pg to 500 ng
Water, nuclease-free	to 10 μ L
Total volume	10 μ L

Table4										
	A	B	C	D	E	F	G	H	I	J
1	Sample	RNA conc (ng/ul)	A260/280	A260/230	RNA amount for 10 uL	minimum amount of RNA	volume for minimum RNA	Water up to 12 uL		
2	5319 hypo 1h	21.7	2.1	2.22	217	30	1.4	8.6		
3	5319 hypo 24h	13.9	2.09	0.28	139	30	2.2	7.8		
4	5319 2.5mM	17.7	2.02	2.09	177	30	1.7	8.3		
5	4517 hypo 1h	4.2	1.67	1.22	42	30	7.1	2.9		
6	4517 hypo 24h	3.3	1.67	0.12	33	30	9.1	0.9		
7	4517 2.5mM	3	1.63	0.02	30	30	10.0	0.0		

Table3				
	A	B	C	D
1	reagent	volume (uL/rxn)	# of rxn	total volume
2	10x dsDNase Buffer	1	12	13
3	dsDNase	1	12	13

2. Mix gently and centrifuge.
3. Incubate at 37°C in PCR machine (or water bath) for 2 minutes.
4. Chill on ice, briefly centrifuge and place on ice.
5. prepare a master mix on ice: (modified the volume of water since we have 12 uL from the last step)

Table5				
	A	B	C	D
1	reagent	volume (uL/rxn)	# of rxn	total volume
2	Maxima cDNA H Minus Synthesis Master Mix (5X)	4	12	52
3	Water, nuclease-free	4	12	52

6. Mix gently and centrifuge.
7. Set up PCR program:

Table6		
	A	B
1	temp	time
2	25	10
3	50	15
4	85	5

8. The reaction product of the first strand cDNA synthesis can be used directly in qPCR. Store at –20°C for up to one week, or –70°C for long-term storage. Avoid freeze-thaw cycles of the cDNA.

QPCR

1.